Reproducible Benchmarking of Tree Ensembles on Tabular Data
A Demonstration Paper for ReproGuard AI

Abstract
This study evaluates reproducible benchmarking practices for tabular classification. We use a public benchmark dataset and compare a Random Forest baseline with Gradient Boosting. The objective is to document enough experimental detail for an independent researcher to repeat the analysis.

1. Dataset and preprocessing
The dataset contains 1,000 labeled samples and 20 features. Missing values are imputed using the median. Numeric features are standardized after the train/test split. We use a stratified 80/20 training/test split and five-fold cross-validation on the training set.

2. Models and hyperparameters
The Random Forest uses n_estimators=200, max_depth=12, and random_state=42. The Gradient Boosting model uses 150 estimators and learning rate=0.05. All experiments use random seed 42.

3. Evaluation
We report accuracy, precision, recall, F1-score, and ROC-AUC. Results are compared with the Random Forest baseline. Each configuration is evaluated in five repeated runs and we report mean and standard deviation.

4. Environment and code availability

Experiments were run with Python 3.12 on a CPU workstation with 16 GB RAM. The reference implementation uses scikit-learn. A public software repository is referenced at https://github.com/scikit-learn/scikit-learn .

5. Limitations

This demonstration is intentionally small and is not intended to establish a scientific claim. It exists to exercise the audit pipeline.

References

[1] Breiman, L. Random Forests. Machine Learning 45, 5-32 (2001). https://doi.org/10.1023/A:1010933404324

[2] Chen, T. and Guestrin, C. XGBoost: A Scalable Tree Boosting System. KDD (2016). https://doi.org/10.1145/2939672.2939785